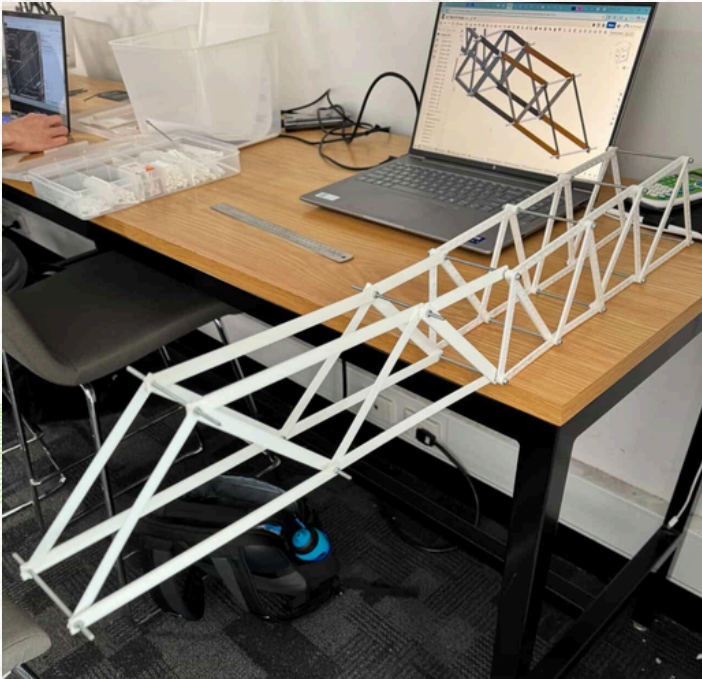


TRUSS DESIGN

Student project - Charlie Truscott



Project Overview

The goal was in your team of 4 to make a Meccano truss (Section A) joined to a 3D-printed cantilever (Section B), designed to hold 2 kg and fail before 2.5 kg, with a span over 400 mm, maximising the span-to-weight ratio. This project involved the use of CAD to design section B while would be assembled to section A once printed

Teamwork and communication

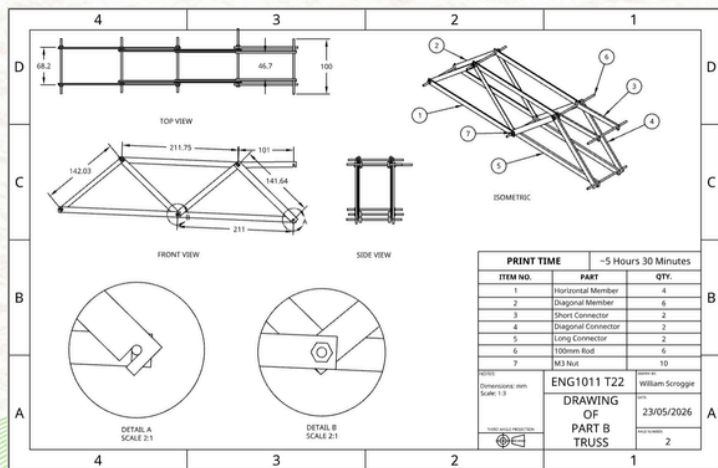
In our team of four we each developed and pitched an individual truss concept before converging on a shared design, giving and receiving feedback on member dimensions, joining mechanisms, and printing settings along the way. Individual statements describe strengthened communication, time management, and conflict-resolution skills from coordinating design decisions and deadlines as a group of four.



(One of our teammates missed the final testing)

TRUSS DESIGN

Student project - Charlie Truscott



Design Process (CAD & 3D Printing)

the final truss was modelled in CAD (isometric, front, side and top views with full dimensioning) before being 3D printed in PLA, with a bill of quantities and print-time estimate (~5.5 hours) documented for Part B. The team learned that CAD modelling and physical construction diverge in practice nut/rod joints don't tighten to a perfect 1:1 fit, which affects the truss's effective member length and real-world performance versus the theoretical design.



Calculations & Testing

Method of joints and sections were applied by hand to solve reaction forces and member forces, followed by factor-of-safety checks identifying critical members. The physical prototype was then tested against these predictions: The physical held 1.5 kg and buckled at 2.0kg, and failed at the 2.5 kg achieving us full marks for our designed target. However our calculations buckled at 1.8 kg. Prompting a discussion of where the theoretical and physical results diverged and how the calculations could be refined.

